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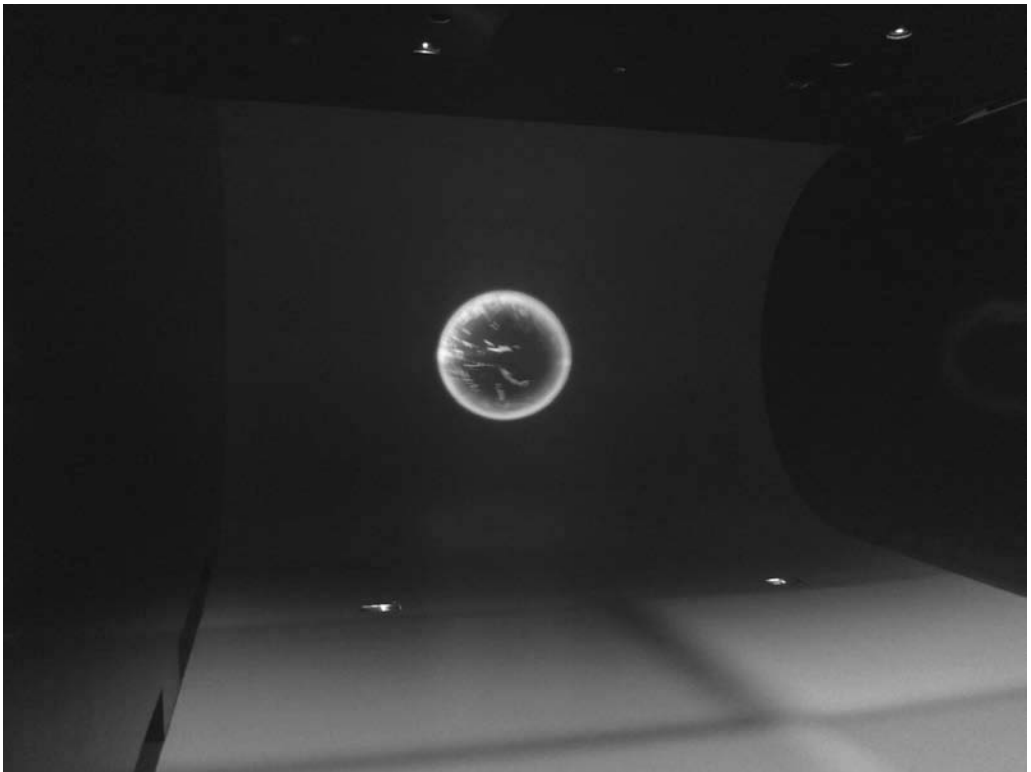


Figure C.1. *Forces of Change* exhibit, Siemens Crystal. A video documenting key planetary stressors to be monitored and managed. Photograph by author.

Planetary Computerization, Revisited

IN TRAVERSING THE AREAS OF WILD SENSING, pollution sensing, and urban sensing, I have sought to build up an account of the environments, entities, and relations that concreate within and through computational sensors. The environmental-sensing projects covered here are situated within a wider array of sensing activities that span from monitoring tides to airflow, as well as observing vegetation and soil conditions. Moving across experimental forests and webcams, migrating animals and climate change, garbage patches and urban air pollution, smart cities, idiotic participation, and automated digital infrastructures, I have discussed how these versions of programmable earths are not singular entities but rather involve the in-forming and multiple activations of distinct techno-geographical environments.

The *becoming environmental of computation* has been the central concept that I have developed to draw out the multiple ways in which earths and environments are programmed and the new entities and experiences that materialize through these processes. But this concept has not been put forward as a way to spatially locate sensor-based media nor to argue for the revolutionary environmental engagements that computational sensors seem to offer. Instead, I have attended to the ways in which sensors become environmental through exchanges and individuations of energy, materialities, subjects of experience, relations, and milieus. Environments are an active part of how actual entities come to concreate, how organisms gain a foothold and endure, and how values are articulated through the unfolding of technical objects. Returning to the ideas put forward in the introduction to this study, we find that programmable earths generate and materialize very particular computable entities, relations, and environments. This is not a simple project of turning the earth into an object or artifact of study, however, but rather a multiply realized set of practices for identifying environments and

processes that can be parsed as sensor data, gathered together into analyzable data sets, and operationalized into new formations of environmental engagement and experience.

This could, of course, have been a cautionary tale of how the earth is becoming a highly instrumented control space, where every activity and environment is under observation. While the capacities of environmental sensors to enable an even further degree of surveillance are no doubt formidable, I have sought to develop a slightly different line of inquiry by attending to the ways in which these computational sensor technologies become environmental to create new subjects of experience. The subjects that would be monitored and would undertake monitoring are not static entities but instead are connected up with generative processes for experiencing environments and other entities in distinct ways. This phenomenon further indicates that how environments are felt and acted upon, whether for political or interventionary purposes, is closely tied to the technologies and subjects that apprehend or feel environments and have concern for environments.

These environmental–computational operations play out differently across the situations that I have discussed throughout *Program Earth*. Wild sensing draws attention to the ways in which we might remotely monitor environmental processes seemingly in absence of human intervention and toward a greater understanding of the real-time ecological relations that unfold. Yet this section also emphasizes that sensing practices entangle milieus, problems, modes of sensing, and matters of concern with particular formations of environmental study, citizen sensing, and engagement. Pollution sensing draws out the often proxy modes of sensing that organisms and societies of objects express and the entities and relations that proliferate within changing milieus and environments. It considers the new creatures and practices of sensing that condescend through the monitoring and collection of pollution data. And urban sensing attends to speculative and implemented smart city scenarios and technologies to consider how sensing operations attempt to make urban spaces and urban citizens into more manageable, efficient, and responsive entities, as well as the ways in which these programs do not always go according to plan.

Sensing, as each of these sections and chapters indicates, is a practice and technological relation that does not simply detect external stimuli to be processed and turned into manageable content. The program earths and programmed participation that condescend here are less linear and substantialist than this. Instead, these programmed environments draw attention to the tunings and attunings that occur through computational and environmental operations. Ways of feeling and accounting for environments and environmental relations are activated or otherwise delimited through computational sensors, and it is these ways of feeling and the practices and subjects that they sustain that I have sought to

make more evident. Environments are not then simply a map of a territory but are a field of resonance and relation that can be drawn into and materialized in the experiences of subjects, whether those subjects are citizen-sensors, soil sensors, moss cams, migrating storks, marine debris in sensor-mapped ocean currents, air pollution-sensing devices, smart buildings, digital infrastructure, or any of the many other actual computational concretizations of environmental-sensing subject-superjects.

Planetary computerization is unfolding apace. As Guattari has suggested (and as was also discussed in the introduction), such computerization might even provide an opportunity for polyphonic engagements across machines, subjects, temporalities, and materialities. Yet while such planetary and computational concrescences are underway, it remains an open question as to whether such computerization might activate the expansive new subjectivities, “creative enchantments,” and imaginaries that Guattari had hoped for.¹ As Guattari writes in characteristically profuse form,

The question that returns here in a haunting fashion is to know why the immense processual potentialities carried by all these computational, telematic, robotic, bureaucratic, biotechnological revolutions so far still only result in a reinforcement of previous systems of alienation, an oppressive mass-mediatization, infantilizing consensual politics. What will enable them finally to lead to a postmedia era, setting them free from segregationalist capitalist values and giving a full lease of life to the beginnings of a revolution in intelligence, sensibility and creation?²

Guattari invested a certain speculative energy into the capacities of new technologies. Yet if the current developments related to environmental sensors and the wider landscape of the Internet of Things are anything to go by, then the “reinforcement of previous systems” seems to be the more likely scenario than a transformation in intelligence, sensibility, or creation. Energy meters and security cameras are pervasive, if less than enchanting, technologies in the ever-connected, always-on, searchable and brand-able big-data predictive worlds in the making. In some ways, this is a logical continuation of the seemingly helpful ubiquitous computing world that Weiser imagined:

When almost every object either contains a computer or can have a tab attached to it, attaining information will be trivial: “Who made that dress? Are there any more in the store? What was the name of the designer of that suit I liked last week?” The computing environment knows the suit you looked at for a long time last week because it knows both of your locations, and it can retroactively find the designer’s name even though that information did not interest you at the time.³

In a contemporary context, where these once-imaginary computing scenarios have now become standard procedures, such engagements sound potentially less enchanting and more in service of tracking a customer-citizen whose attention is perpetually tuned to consumption.

In this respect, it is also useful to return to Simondon's critique of automation (and by extension, first-order cybernetics) as discussed in the last chapter, where he suggests that automation is quite a limited and uninteresting way to mobilize technology. Instead, Simondon suggests that if we attend to the values articulated and transduced through technology, we might be able to identify opportunities for active experimentation within these cultural and machinic registers as expressions of relationality—and even, potentially, equality and invention. Open technology was a focus of Simondon's in this regard, not in the usual sense of openness that circulates today in relation to “free” hardware and software, but rather an openness that consisted of experimentation with the entities, relations, and collectives that might be individuated through technological engagements.

Such an approach also points toward considerations of how not to fix technology into conditions of inequality but instead to open it into encounters that are more-than-technical encounters with technocultural creativity and responsibility. This would require going beyond engagements with technology as a utensil or instrument, as mentioned in the previous chapter.⁴ It would also require going beyond engagements with environments as computational problems to be solved. Instead of remaining firmly embedded within the utensil-problem space, where environmental sensors facilitate increasingly regimented and automated ways of life, another approach might be to consider how environmental computational practices open into experimentation, expanded experiences, and speculative adventures.

PROGRAMMING EARTHS, EXPERIMENTING WORLDS

The problem of automation is closely tied to the processes and imaginaries of programming. Programming, as this work has suggested, is not simply an automatic code or script foisted on the world through a set of discursive enactments. The usual sense of the programmatic often refers exactly to those scripted and less-than-inventive engagements. And so there proliferate antiprograms and counterprograms, seemingly as a way to break out of programs of control. In relation to a discussion of “programmatic statements,” Massumi has suggested that breaking with such statements is a way to generate encounters with politics.⁵ Programmatic statements might at first appear to be expressions of politics—of a “correct” position to be taken in relation to a certain problematic. But these can actually reinforce deadening certainties that prevent a more engaged and engaging encounter with matters of concern. The point of politics is to go beyond programmatic statements in order to open our inhabitations to more speculative

encounters. It is through a break with programmability—both as an automatic code and as a correct position—vis-à-vis experimentation that political inventiveness might unfold.

Programmability, as I have suggested throughout this study, is also a somewhat fraught process, verging on the enactment of scripts for certain outcomes but giving rise to indeterminate environments, entities, and relations. Programmability in the context of program earths is continually unfolding the complex if at times troubling attempts to make environments into spaces of observation, distributed experience, and even automated management. Programming might then be addressed as a technical process that is involved in individuating entities, relations, and milieus. These processes are not a mere list of entities, but rather describe the *collective* articulation of potential, as well as all that is more-than-human in human engagements with technologies.

Participation then becomes an important topic of consideration when attending to the operations of environmental sensors. The citizen-sensing practices that I have discussed throughout this study have demonstrated not just how citizenship, participation, and practices of engaging with environmental problems are bound up with and individuated through environmental sensors, but also how the “citizen” in citizen sensing is an entity that shifts and expands to include other practices constitutive of citizenship, and which might include more ecological ways of accounting for how citizenly engagements unfold. In this way, I have not sought to reinscribe the somewhat pervasive notion that citizen-sensing technologies are liberatory tools for achieving more democratic engagement, or that a heightened ability to work with computational sensors might convey an increased amount of political capability to those few skilled citizens.

Instead, what I am more inclined to consider is that, in the current circumstances, increasingly one is either in the network or out.⁶ While Internet dark rooms, off-grid spaces, and antique (read: nonsmart) appliances might be one strategy within an “only dis/connect” approach to the modes of planetary computerization that are now unfolding, such strategies do not realize another space *within* networks so much as opt out of these computational relations altogether. In relation to milieus, both Simondon and Foucault articulate, in a transformed way, Canguilhem’s notion that to be simply *in opposition* is to be already defeated. Living entities must extend themselves and make something of their milieus.⁷ In this way, Foucault attended to power not by making simple exhortations to resist its influence but rather by noting that power is a force that runs through ways of life, and that it was necessary to reroute power where possible to invent more creative and engaging ways of life. Simondon’s point was that a more thoroughgoing engagement with technical objects might materialize through attempts to experiment with other relations with technology, rather than to opt out of these engagements altogether. Technology is not simply to be dismissed as other-than-human,

Simondon has also argued, but rather is part of the collective potential that makes us human in particular ways as it individuates entities and our possibilities of relating. This in turn also has consequences for the extensions and exchanges made with environments and more-than-human organisms.

In drawing out this discussion of how sensor technologies transindividuate environments and entities, individuals, and collectives, I have attempted to build a context in which the *power* of speculation might also become more apparent. This power unfolds through the ways in which propositions are made: What are the computational environments we would inhabit, how do they lure us into becoming together, and what are the processes and practices that give rise to these concrescences? This power also takes hold through the ability to undertake persuasive practices, to infect and in-form the conditions and entities that might activate speculative engagements. The power of speculation might also be articulated across the moments and sites of concretization, where abstract technology takes root, becomes *more indeterminate*, and so potentially is generative of multiple speculations.

In this respect, speculation is as much about propositioning, instigating, and triggering—beyond the usual automated sensor-actuator triggers of cybernetics—toward indeterminacy and openness. As Whitehead has suggested, a speculative project is most interesting when it is involved in “bringing adventures into existence.” Rather than demystifying, mapping out, and nailing down everything in a grand gesture of rationalism,⁸ adventure seeks to create conditions for both hope and change.⁹ A speculative adventure is then an approach invested in experimenting worlds. Less a matter of polemics, adventure and speculation are about making particular things matter—of generating environments and entities that are able to take hold in particular ways because they have exercised power through persuasion (and experience).¹⁰

Speculative approaches to research and practice are emerging across multiple fields as a way to develop not simply descriptive engagements with topics but also propositions that invent new possibilities for research and practice. One mode of experimentality developed throughout *Program Earth* has involved encountering environmental sensing as a series of propositions for considering how the becoming environmental of computation offers up opportunities for creative and practical as much as analytical engagements.

Citizen-sensing projects often attempt to find ways to broaden the scope of observational practices beyond the sciences exclusively and to make ecological observation more accessible and engaging to a diverse range of participants. Such citizen-sensing projects intend to democratize the collection and use of environmental sensor data in order to increase citizen engagement in environmental issues. The process of gathering and making these observations more participatory is often a way to overcome the relative crisis of environmental engagement

in political and cultural spheres: by making environmental change more evident and distributed across sensing subjects, environmental action may also be facilitated. But these practices also tune in subjects, environments, technologies, and multiple other entities into shared registers of sense making.

Program Earth has considered how or whether environmental-sensing and citizen-sensing practices enable expanded engagement not just with the “message” of environmental change but also with understanding environmental concerns at a more intimate level. Through discussing the rise and proliferation of environmental sensing practices within scientific, creative, citizen, and urban applications, I have analyzed the distinct practices of environmental politics that congregate in relation to these technologies, where citizens make use of technoscientific devices at times to reroute the usual spaces of environmental engagement and expertise.

An obvious observation to make would be that citizenship is performed through these sensing technologies. But how does this mode of citizenly practice square with the stabilizations of citizenship to which these practices might refer and/or resonate (or dissonate) within democratic contexts? A citizen is not newly emergent with every use of sensors, but these technologies are involved in re-articulating and recasting the materiality, spaces, practices, collectivities, infrastructures, imaginings, abstractions, processes—in other words—concretizations of citizenship. From a perspective informed by Whitehead and Simondon, we could say that citizenship is historically immanent and processual, as well as a site of dynamically articulated collective potential. We could also say that a citizen is not an exclusively human-based subject of experience but inevitably is also part of an extended ecology of attachments.

Citizen sensing, as it is typically conceived, is often positioned as somewhat continuous with citizen science, an activity that might be augmenting science but through more digitally enabled devices. And yet, citizen sensing might also be an expanded way of practicing environmental sensing in relation to pedagogical and political aims.¹¹ The development of these alternative environmental monitoring practices might, on one level, focus on technologies for engaging with environments, where alternative monitoring practices might be a way to question official versions of events, while also developing tools for continually engaging with environmental issues. Yet, on another level, as practices situated within and generative of milieus, citizen-sensing practices are also inevitably techno-geographical as well as speculative. If more thoroughly engaged with the environmental concerns they would monitor, they might even generate new subjects and politics of experience. Citizen-sensing practices might then not necessarily be data-collection exercises but rather ways of making particular environments and environmental concerns matter and gain a foothold. In this way, citizen sensing might move beyond the unquestioned if rather problematic equation of data-equals-action to

engage with the extended milieus, moments of resonance, and points of collective potential that might be forceful sites of investigation and practice.

In discussing environmental sensing and citizen-sensing practices, I have attended to the experimental aspects of these engagements in thinking about the distinct human and more-than-human distributions of experience that are brought together and which take on consistency. However, rather than talk *about* these practices through a discursive analysis that treats them as an object of study, I have considered how an approach attuned to experimentation also seeks to mobilize and invent abstractions along with concurring practices, materialities, and environments.¹² In other words, I have attempted to write toward the edge of the page, in-between the spaces of thinking and practicing, such that these are not seen as oppositional modalities but rather as continuous in their commitment to experimenting with ways of encountering environments and environmental matters of concern with creative, critical, and political a/effect.

This research finally suggests that new approaches to computation might be developed where digital devices expand beyond automated and user-controlled applications toward more speculative and “open” engagements. One question to bring to any environmental sensing project might then be: How does it give rise to speculative adventures, or otherwise prevent experimentations with worlds? The becoming environmental of computation might be as much a proposition as a fact. Sensorized environments are propositions for particular types of worlds to take hold and for distinct subjects of experience to be sustained. Sensorized environments and citizen-sensing practices put in motion specific ways of feeling worlds and of making particular problems matter. In this sense, citizen-sensing practices materialize not as easy fixes to making environmental engagement more democratic but rather as particular *expressions* of environmental problems, politics, and citizenship. Given that these projects are so actively making worlds, they are also sites that might then be encountered through further speculation.

Simondon might have opted to use the term “invention” over and above “speculation,” and at the same time he investigated how processes of transindividuation always left a preindividual reserve that was a site for further potentialities.¹³ And yet, invention is also a topic and practice that runs through Stengers and Whitehead. Stengers discusses how speculation involves attending to the “invention of the field in which the problem finds its solution.”¹⁴ There is an adventure that is undertaken in formulating questions, and concepts can transform “the way in which a situation raises a problem.”¹⁵ A speculative approach can transform questions and the character of experience—it is a conceptual approach that is charged with its ability to shake up and recast the usual approaches to problems.¹⁶

I end this discussion by considering the theoretical influence of Whitehead and Stengers, who, in a related way, discuss how it might be possible to be *for a*

world, and not simply of the world.¹⁷ Our inhabitations are ways of experiencing and, in turn, making worlds. The etho-ecological relations that we articulate along with multiple others require certain environments to take hold and endure in order for those relations and experiences to be sustained. This is a process of persuasion, infection, and power. It is also a process of experimentation and ethics. The milieus that are put into play through environmental sensors are ecologies of amplification—as they connect, they intensify.¹⁸ As computational sensors increasingly take hold and concreate entities, relations, and environments in distinct ways, the question of what sorts of worlds—or program earths—we are involved in sustaining comes into play. While this is not a simple proposal to adopt one particular relation to these technologies, this study suggests that the world-making operations and experiences of sensors might become an area of more intensive—if openly indeterminate—engagement. *Program Earth* is one incomplete and speculative attempt at making an opening into this space.

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